

● MEDIUM

● ALGEBRA

● ANSWER KEY

Algebra

10

10 answers · Rationale-rich · Teach from doc

Q1**.1764**

Accept also: .1765, 3/17

The correct answer is $\frac{3}{17}$. It's given that line j is perpendicular to line k in the xy -plane. This means that the slope of line j is the negative reciprocal of the slope of line k . The equation of line k , $y = -\frac{17}{3}x + 5$, is written in slope-intercept form $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept of the line. It follows that the slope of line k is $-\frac{17}{3}$. The negative reciprocal of a number is -1 divided by the number. Therefore, the negative reciprocal of $-\frac{17}{3}$ is $\frac{-1}{-\frac{17}{3}}$, or $\frac{3}{17}$. Thus, the slope of line j is $\frac{3}{17}$. Note that 3/17, .1764, .1765, and 0.176 are examples of ways to enter a correct answer.

Q2**D****D. 16**

Choice D is correct. The value of $4 - 3x$ can be found by isolating this expression in the given equation. Subtracting 2 from both sides of the given equation yields $94 - 3x = 84 - 3x + 16$. Subtracting $84 - 3x$ from both sides of this equation yields $94 - 3x - 84 - 3x = 16$, which gives $14 - 3x = 16$, or $4 - 3x = 16$. Therefore, the value of $4 - 3x$ is 16.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not $4 - 3x$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3**C**

C. $f(x) = -\frac{4}{3}x + \frac{1}{3}$

Choice C is correct. A linear function can be written in the form $f(x) = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept of the line. The slope of the graph can be found using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the values of the given points into this formula yields $m = \frac{-5 - 3}{4 - (-2)}$ or $m = \frac{-8}{6}$, which simplifies to $m = -\frac{4}{3}$. Only choice C shows an equation with this slope.

Choices A, B, and D are incorrect and may result from computation errors or misinterpreting the given information.

Q4**7**

The correct answer is 7. Subtracting the second equation from the first equation eliminates the variable y .

$$\begin{array}{rcl} 5x + 3y & = & 38 \\ -(x + 3y & = & 10) \\ \hline 4x & = & 28 \end{array}$$

Dividing both sides of the resulting equation by 4 yields $x = 7$.

Q5**A****A.** 34

Choice A is correct. It's given that the truck can tow a trailer if the combined weight of the trailer and the boxes it contains is no more than 4,600 pounds. If the trailer has a weight of 500 pounds and each box weighs 120 pounds, the expression $500 + 120b$, where b is the number of boxes, gives the combined weight of the trailer and the boxes. Since the combined weight must be no more than 4,600 pounds, the possible numbers of boxes the truck can tow are given by the inequality $500 + 120b \leq 4,600$. Subtracting 500 from both sides of this inequality yields $120b \leq 4,100$. Dividing both sides of this inequality by 120 yields $b \leq \frac{205}{6}$, or b is less than or equal to approximately 34.17. Since the number of boxes, b , must be a whole number, the maximum number of boxes the truck can tow is the greatest whole number less than 34.17, which is 34.

Choice B is incorrect. Towing the trailer and 35 boxes would yield a combined weight of 4,700 pounds, which is greater than 4,600 pounds.

Choice C is incorrect. Towing the trailer and 38 boxes would yield a combined weight of 5,060 pounds, which is greater than 4,600 pounds.

Choice D is incorrect. Towing the trailer and 39 boxes would yield a combined weight of 5,180 pounds, which is greater than 4,600 pounds.

Q6**4.51**

Accept also: 451/100

The correct answer is 4.51. It's given that the equation $4.51x + 6.07y = 896.86$ represents this situation, where x is the number of smaller containers sold, y is the number of larger containers sold, and 896.86 is the store's total sales, in dollars, of blueberries last month. Therefore, $4.51x$ represents the store's sales, in dollars, of smaller containers, and $6.07y$ represents the store's sales, in dollars, of larger containers. Since x is the number of smaller containers sold, the price, in dollars, of each smaller container is 4.51.

Q7**C**

C. $950 + 50(t - 2) = 1,100$

Choice C is correct. It's given that the cost to rent a commercial fishing boat is \$ 950 for the first 2 hours and an additional \$ 50 per hour for each hour after the first 2 hours. It's also given that t represents the total number of hours and $t > 2$. Therefore, the number of additional hours after the first 2 hours can be represented with the expression $t - 2$. The cost for these additional hours is \$ 50 per hour, so the cost for the additional hours can be represented by the expression $50t - 2$. The total cost can be calculated by adding the cost for the first 2 hours to the cost for the additional hours and can be represented by the expression $950 + 50t - 2$. It's also given that the total cost to rent the commercial fishing boat from the company for t hours is \$ 1,100. Thus, the equation that represents this situation is $950 + 50t - 2 = 1,100$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q8**D**

D. $h(x) = -\frac{7}{2}x + 113$

Choice D is correct. An equation defining h can be written in the form $y = mx + b$, where $y = hx$, m represents the slope of the graph of $y = hx$ in the xy -plane, and b represents the y -coordinate of the y -intercept of the graph. It's given that $h28 = 15$ and $h26 = 22$. It follows that the points 28, 15 and 26, 22 are on the graph of $y = hx$ in the xy -plane. The slope can be found by using any two points, x_1, y_1 and x_2, y_2 , and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting 28, 15 and 26, 22 for x_1, y_1 and x_2, y_2 , respectively, in the slope formula yields $m = \frac{22 - 15}{26 - 28}$, or $m = -\frac{7}{2}$. Substituting $-\frac{7}{2}$ for m and 28, 15 for x, y in the equation $y = mx + b$ yields $15 = -\frac{7}{2}28 + b$, or $15 = -98 + b$. Adding 98 to both sides of this equation yields $113 = b$. Substituting $-\frac{7}{2}$ for m and 113 for b in the equation $y = mx + b$ yields $y = -\frac{7}{2}x + 113$. Since $y = hx$, it follows that the equation that defines h is $hx = -\frac{7}{2}x + 113$.

Choice A is incorrect. For this function, $h26 = \frac{109}{7}$, not $h26 = 22$.

Choice B is incorrect. For this function, $h28 = 105$, not $h28 = 15$, and $h26 = \frac{739}{7}$, not $h26 = 22$.

Choice C is incorrect. For this function, $h28 = -75$, not $h28 = 15$, and $h26 = -68$, not $h26 = 22$.

Q9**C**

C. $m + v = 60$

$$0.75m + 0.55v = 39$$

Choice C is correct. It's given that m represents the original price, in dollars, of the mirror, and v represents the original price, in dollars, of the vase. It's also given that the combined original price for the mirror and the vase is \$ 60. This can be represented by the equation $m + v = 60$. After a 25% discount to the mirror is applied, the sale price of the mirror is 75% of its original price. This can be represented by the expression $0.75m$. After a 45% discount to the vase is applied, the sale price of the vase is 55% of its original price. This can be represented by the expression $0.55v$. It's given that the combined sale price for the two items is \$ 39. This can be represented by the equation $0.75m + 0.55v = 39$. Therefore, the system of equations consisting of the equations $m + v = 60$ and $0.75m + 0.55v = 39$ gives the original price m , in dollars, of the mirror and the original price v , in dollars, of the vase.

Choice A is incorrect. The second equation in this system of equations represents a 45% discount to the mirror and a 25% discount to the vase.

Choice B is incorrect. The second equation in this system of equations represents a 55% discount to the mirror and a 75% discount to the vase.

Choice D is incorrect. The second equation in this system of equations represents a 75% discount to the mirror and a 55% discount to the vase.

Q10**D**

D. $0.6 \leq s \leq 1.8$

Choice D is correct. It's given that for a snowstorm in a certain town, the minimum rate of snowfall recorded was 0.6 inches per hour, the maximum rate of snowfall recorded was 1.8 inches per hour, and s represents a rate of snowfall, in inches per hour, recorded for this snowstorm. It follows that the inequality $0.6 \leq s \leq 1.8$ is true for all values of s .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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